

BuildFlow

Smart Custom PC Builder & Intelligent Order Fulfillment System

Project Proposal Document

(Problem Statement & System Overview)

1. Project Title

BuildFlow: Smart Custom PC Builder & Intelligent Order Fulfillment System

2. Introduction

The rapid growth of custom PC building has increased the demand for platforms that not only help users configure compatible computer systems but also efficiently manage the complex operational processes that occur after an order is placed.

Most existing custom PC websites primarily focus on allowing customers to select compatible hardware components and complete the purchase. However, the backend operations—including inventory reservation, component allocation, assembly management, quality assurance, packaging, shipment tracking, and operational monitoring—are often handled through separate enterprise software systems that customers never interact with.

This separation creates several operational challenges, such as inventory inconsistencies, duplicate processing, order delays, lack of transparency, and inefficient coordination between different departments.

The proposed system, **BuildFlow**, addresses these issues by integrating the complete lifecycle of a custom PC order into a single web application developed using the MERN stack.

Instead of functioning as only an e-commerce website, BuildFlow serves as an end-to-end order fulfillment platform that connects customers, warehouse personnel, assembly technicians, quality assurance teams, logistics staff, and administrators through one centralized system.

3. Problem Statement

Current custom PC builder websites generally stop functioning once the customer completes the payment.

The operational workflow that follows—including inventory management, warehouse coordination, component allocation, PC assembly, testing, packaging, shipping, and order monitoring—is usually managed using multiple disconnected software systems or manual processes.

This creates several real-world problems:

- Inventory overselling due to simultaneous orders.
- Lack of automatic compatibility verification.
- Duplicate execution of operational tasks.
- Poor visibility into order progress.
- Difficulty in tracking operational history.
- Manual coordination between warehouse and assembly teams.
- Inefficient handling of failed operations.
- Limited analytical insights into production workflows.

As businesses scale, managing these processes manually becomes increasingly difficult and error-prone.

Therefore, there is a need for an integrated platform that combines customer-facing PC configuration with enterprise-level order fulfillment and operational management.

4. Proposed Solution

BuildFlow provides a centralized solution that combines both customer and enterprise workflows.

Customers can configure their own custom PCs by selecting hardware components while the system validates compatibility automatically.

Once the order is confirmed, the platform transitions into an intelligent fulfillment engine that manages every stage of the production lifecycle, including inventory allocation, assembly scheduling, quality inspection, packaging, shipment, and delivery tracking.

Each department interacts with the same centralized system using role-based dashboards, ensuring transparency and reducing operational errors.

5. Existing System

Current custom PC builder platforms generally provide:

- Component selection
- Basic compatibility checking
- Shopping cart
- Online payment
- Order placement

After payment, the internal processing is typically managed outside the platform.

Limitations include:

- No inventory reservation
 - No warehouse workflow
 - No assembly tracking
 - No production management
 - No audit logging
 - No duplicate operation prevention
 - Limited order visibility
 - Separate management systems
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6. Proposed System

BuildFlow extends the traditional PC builder by integrating enterprise operational workflows into a single platform.

The system provides:

- Custom PC Builder
- Compatibility Validation Engine

- Inventory Reservation Engine
 - Order Fulfillment Workflow
 - Assembly Management
 - Quality Assurance Tracking
 - Packaging Management
 - Shipment Management
 - Audit Logging
 - Analytics Dashboard
 - Role-Based Access Control
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7. Objectives

The primary objectives of BuildFlow are:

- Simplify custom PC configuration.
 - Prevent incompatible hardware selections.
 - Eliminate inventory conflicts.
 - Automate operational workflows.
 - Improve production transparency.
 - Reduce duplicate operational actions.
 - Maintain complete activity history.
 - Improve warehouse coordination.
 - Enhance customer order tracking.
 - Generate business insights through analytics.
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8. Scope of the Project

The project covers the complete lifecycle of a custom PC order.

It includes:

Customer Module

- Registration/Login
- Component Selection
- Compatibility Validation
- Save Build
- Build Comparison

- Shopping Cart
 - Order Placement
 - Order Tracking
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Warehouse Module

- Inventory Management
 - Stock Reservation
 - Component Allocation
 - Low Stock Alerts
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Assembly Module

- Assembly Queue
 - Technician Assignment
 - Build Status
 - Progress Updates
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Quality Assurance Module

- Testing Reports
 - Quality Verification
 - Pass/Fail Decision
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Logistics Module

- Packaging
 - Shipment Creation
 - Delivery Tracking
 - Failed Shipment Handling
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Administration Module

- User Management
- Component Management

- Analytics
 - Reports
 - System Monitoring
 - Audit Logs
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9. Real-World Workflow

Customer Registration



Build Custom PC



Compatibility Validation



Add to Cart



Checkout



Payment



Inventory Reserved



Components Allocated



Assembly Assigned



Assembly Completed



Quality Testing



Packaging



Shipment



Delivery



Order History

10. Key Features

1. Component Catalog Management

Administrators manage:

- CPUs
- GPUs
- Motherboards
- RAM
- SSD
- HDD
- Power Supply
- Cabinets
- Cooling Systems

2. Compatibility Validation Engine

Automatically validates:

- CPU Socket
- Motherboard Chipset
- RAM Type
- PSU Wattage
- Storage Interface
- Cabinet Size
- Cooler Support

Users receive compatibility warnings before checkout.

3. Custom Build Management

Users can:

- Save Builds
 - Edit Builds
 - Compare Builds
 - Share Builds
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4. Intelligent Inventory Reservation

Once payment succeeds:

Inventory is automatically reserved.

This prevents:

- Stock conflicts
 - Double allocation
 - Overselling
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5. Order Fulfillment Engine

Every order follows predefined states.

Example:

Order Created



Payment Verified



Inventory Reserved



Assembly



Testing



QC Passed



Packaging



Shipping



Delivered

6. Duplicate Action Prevention

Critical operations execute only once.

Examples:

- Payment confirmation
- Shipment generation
- Inventory reservation
- Assembly completion

7. Permanent Audit Logs

Every operation is recorded.

Example:

Inventory Reserved

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Assembly Started

↓

Assembly Completed

↓

QC Approved

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Shipment Generated

No records are deleted.

8. Automated Retry Mechanism

If external operations fail:

- Shipment API
- Email
- SMS
- Payment confirmation

The system retries automatically before marking the process as failed.

9. Dashboard & Analytics

Provides insights into:

- Daily Orders
 - Revenue
 - Component Sales
 - Inventory Status
 - Build Completion Rate
 - Delivery Performance
 - Warehouse Efficiency
 - Assembly Time
 - Technician Productivity
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11. User Roles

Customer

- Configure PC
 - Place Orders
 - Track Orders
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Warehouse Manager

- Manage Inventory
 - Reserve Components
 - Update Stock
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Assembly Technician

- Assemble Systems
 - Update Build Progress
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QA Engineer

- Test Completed Builds
 - Approve Quality
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Logistics Manager

- Package Orders
 - Dispatch Shipments
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Administrator

- System Configuration
 - User Management
 - Analytics
 - Reports
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12. Functional Requirements

1. Component Catalog Management
2. Custom PC Configuration
3. Compatibility Validation Engine
4. Shopping Cart
5. Secure Authentication
6. Role-Based Access Control
7. Inventory Reservation
8. Order Management
9. State-Based Workflow Engine
10. Assembly Tracking
11. Quality Assurance Management

- 12. Shipment Tracking
 - 13. Dashboard Analytics
 - 14. Audit Logging
 - 15. Duplicate Action Prevention
 - 16. Automated Retry Mechanism
 - 17. Notifications
 - 18. Report Generation
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13. Non-Functional Requirements

Performance

- Response time below 3 seconds for normal operations.

Scalability

- Support increasing users and component catalog without major redesign.

Security

- JWT authentication.
- Password hashing.
- Role-based authorization.
- Input validation.

Reliability

- Prevent duplicate operations.
- Ensure transaction consistency.
- Maintain audit history.

Usability

- Responsive interface.
 - Intuitive dashboards.
 - Easy navigation.
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14. Expected Outcomes

The completed system will:

- Simplify custom PC purchasing.
 - Reduce inventory errors.
 - Improve warehouse coordination.
 - Automate fulfillment processes.
 - Increase operational transparency.
 - Provide real-time order visibility.
 - Generate useful business analytics.
 - Simulate a real-world enterprise workflow suitable for educational and demonstration purposes.
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15. Technologies

Frontend

- React.js
- React Router
- Tailwind CSS / Bootstrap
- Axios

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

Authentication

- JWT
- bcrypt

Additional

- Multer (image uploads)
 - Cloudinary (optional)
 - Socket.IO (real-time updates)
 - Chart.js / Recharts (analytics)
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16. Conclusion

BuildFlow is more than a custom PC builder; it is an **enterprise-grade order fulfillment platform** that models the complete operational lifecycle of a custom PC manufacturing business. By integrating customer interaction, inventory control, production management, quality assurance, logistics, and analytics into a unified MERN application, the project demonstrates practical software engineering concepts such as workflow automation, role-based access control, state-driven order management, and transaction consistency. This makes it a comprehensive and industry-relevant project that is both technically challenging and achievable for a fifth-semester software engineering team.